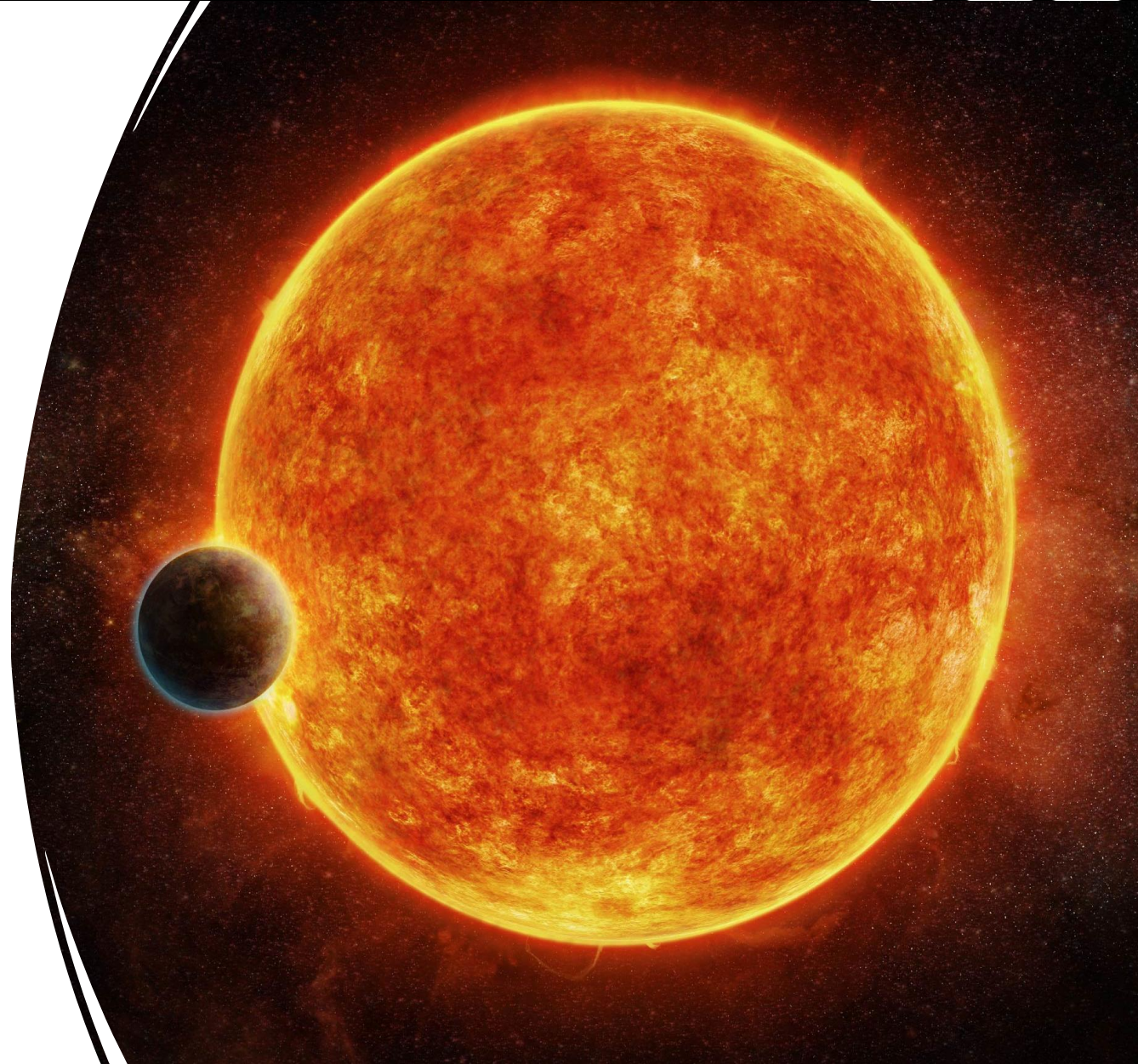


The Effect of Gravity Darkening during Exoplanet Transits

*Constraining the 3D spin-orbit geometry
of HAT-P-70*



Outline of the Project

Test whether **gravity-darkened transit photometry** can constrain the **true 3D spin-orbit angle** in extreme exoplanet systems.



Use **spin-orbit angles** as diagnostics of **planet formation and migration**.



Focus on **HAT-P-70**, a rapidly rotating hot-Jupiter system with strong gravity-darkening and reported misalignment.



Analyse **high-precision TESS photometry** to detect transit asymmetries beyond standard limb-darkened models.



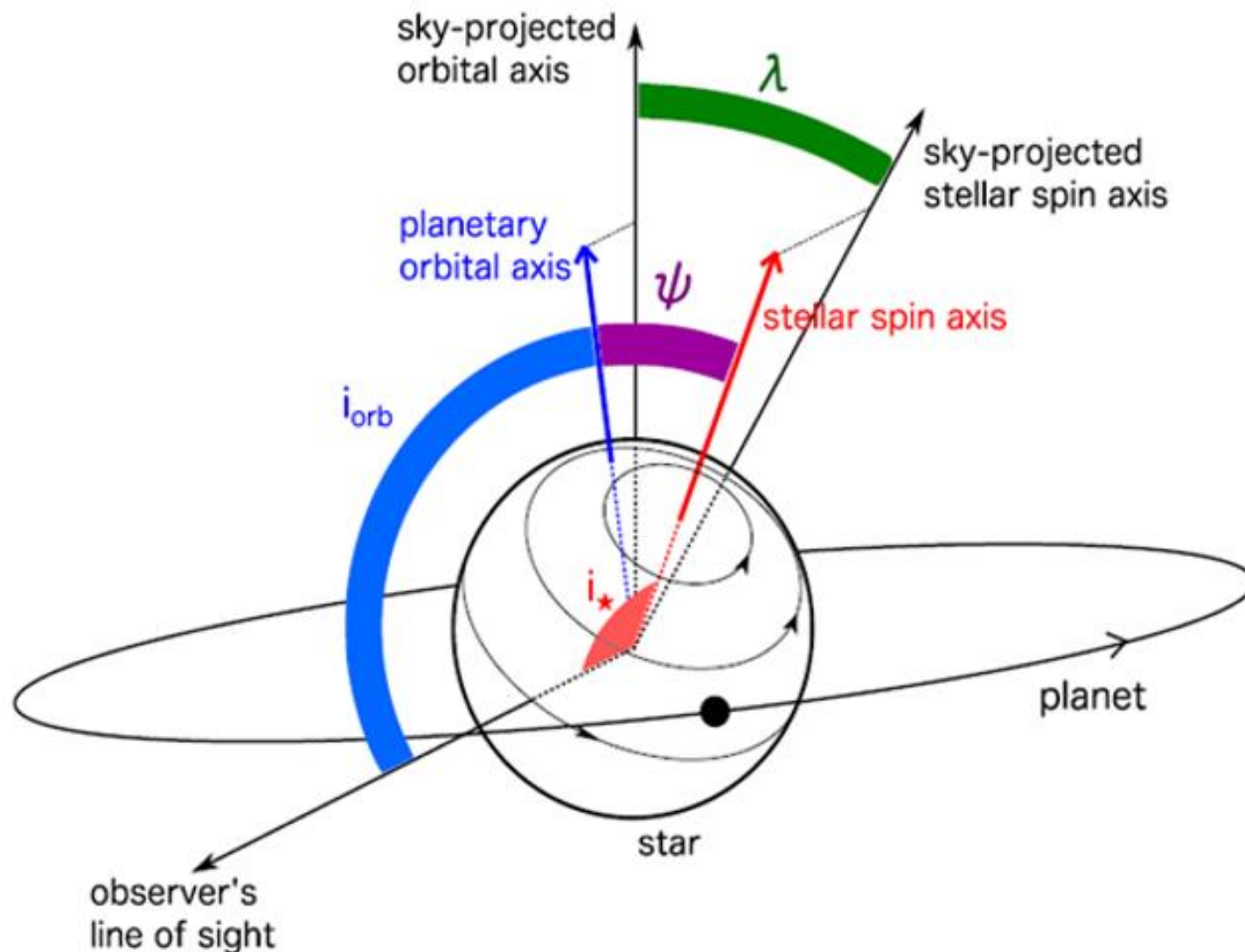
Model these asymmetries with **gravity-darkened transit simulations** and **statistical inference** to constrain the system's 3D geometry.

- The geometry of a transiting exoplanet system is described by several key angles:

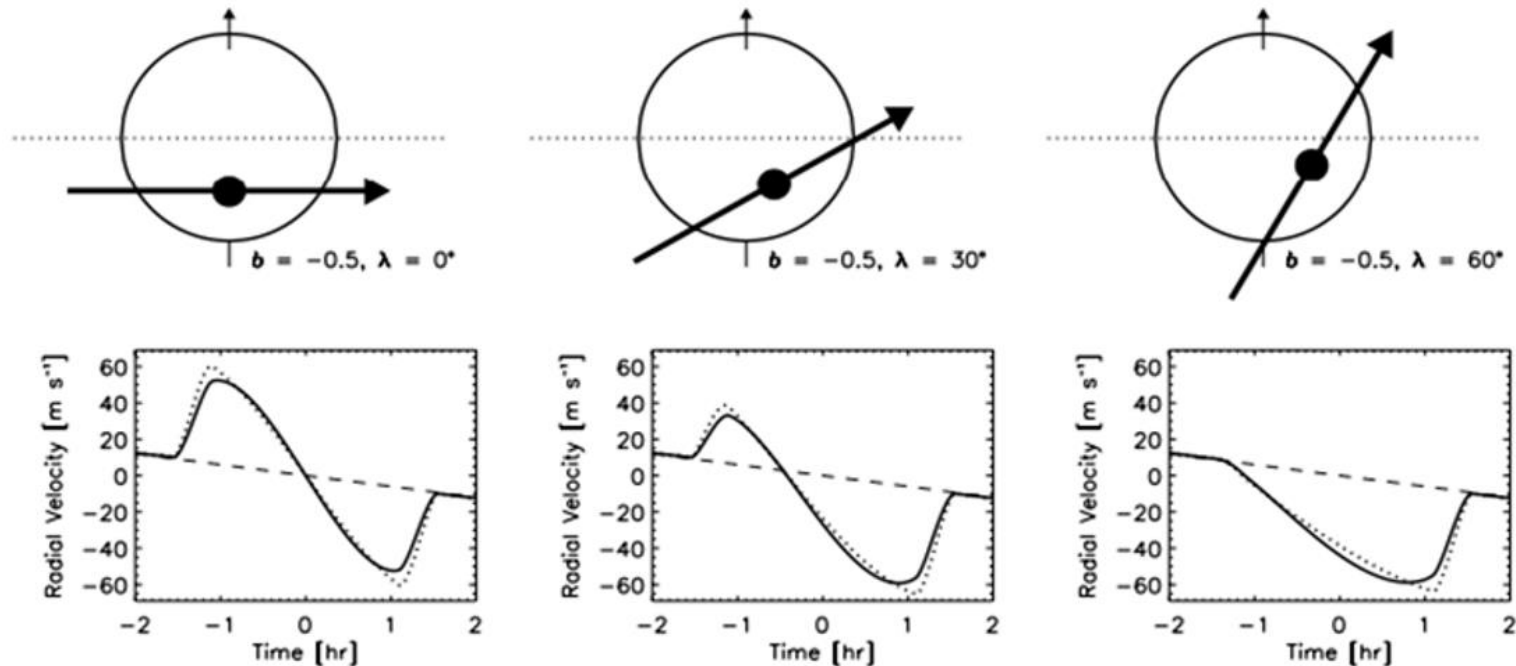
- The orbital inclination, i_{orb}
- The stellar inclination, i_*
- The sky-projected spin-orbit angle, λ

- These quantities define the true three dimensional spin-orbit angle, ψ through this simple equation:

- $$\cos\psi = \cos i_* \cos i_{orb} + \sin i_* \sin i_{orb} \cos\lambda$$

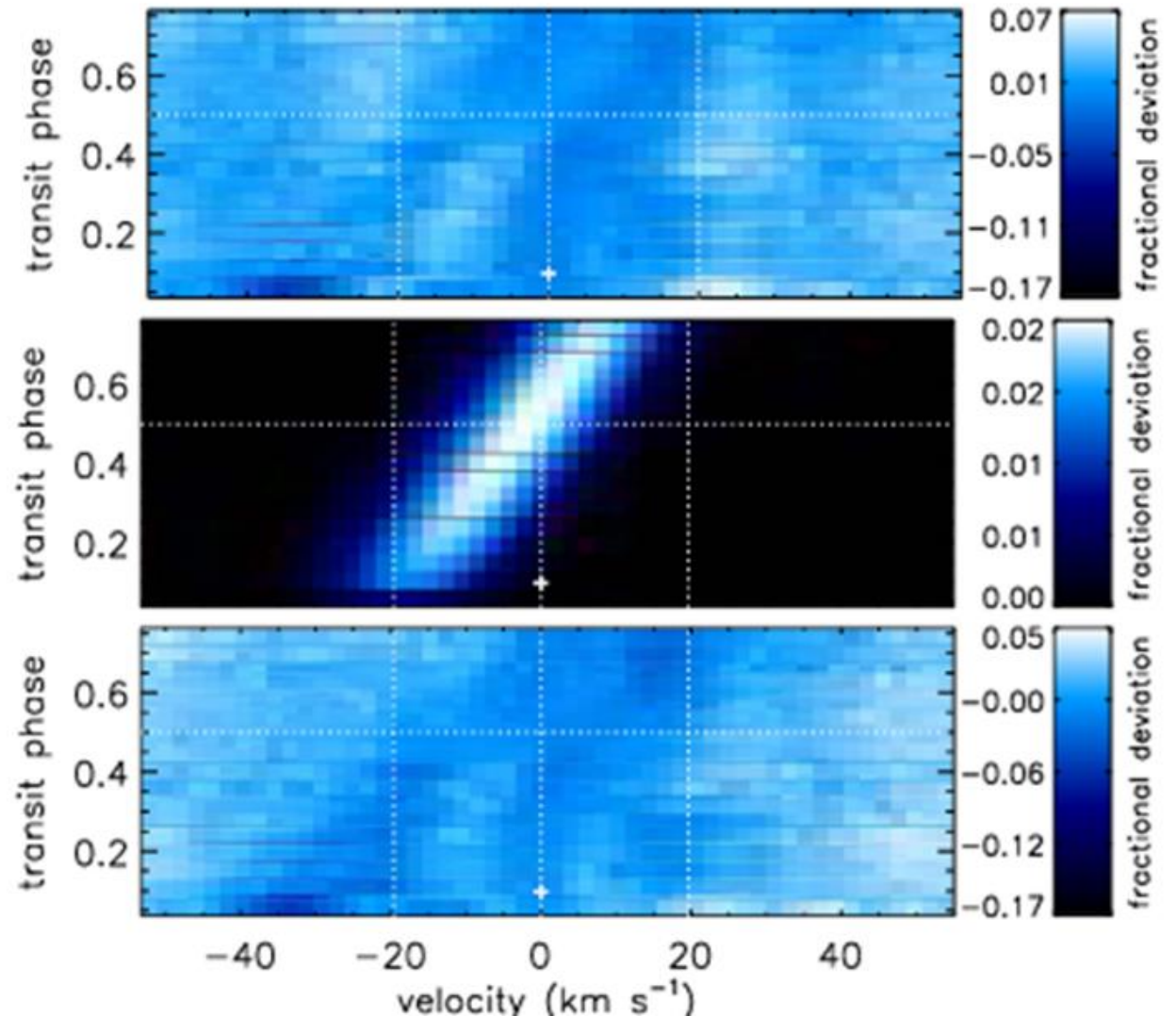


- Spectroscopic techniques such as the Rossiter–McLaughlin effect and Doppler tomography are routinely used to measure projected spin–orbit angle.
- For hot, rapidly rotating stars, severe rotational line broadening significantly degrades radial-velocity precision and limits the reliability of these methods.
- In these cases, the stellar inclination remains largely unconstrained, preventing a full three-dimensional reconstruction of the system geometry.



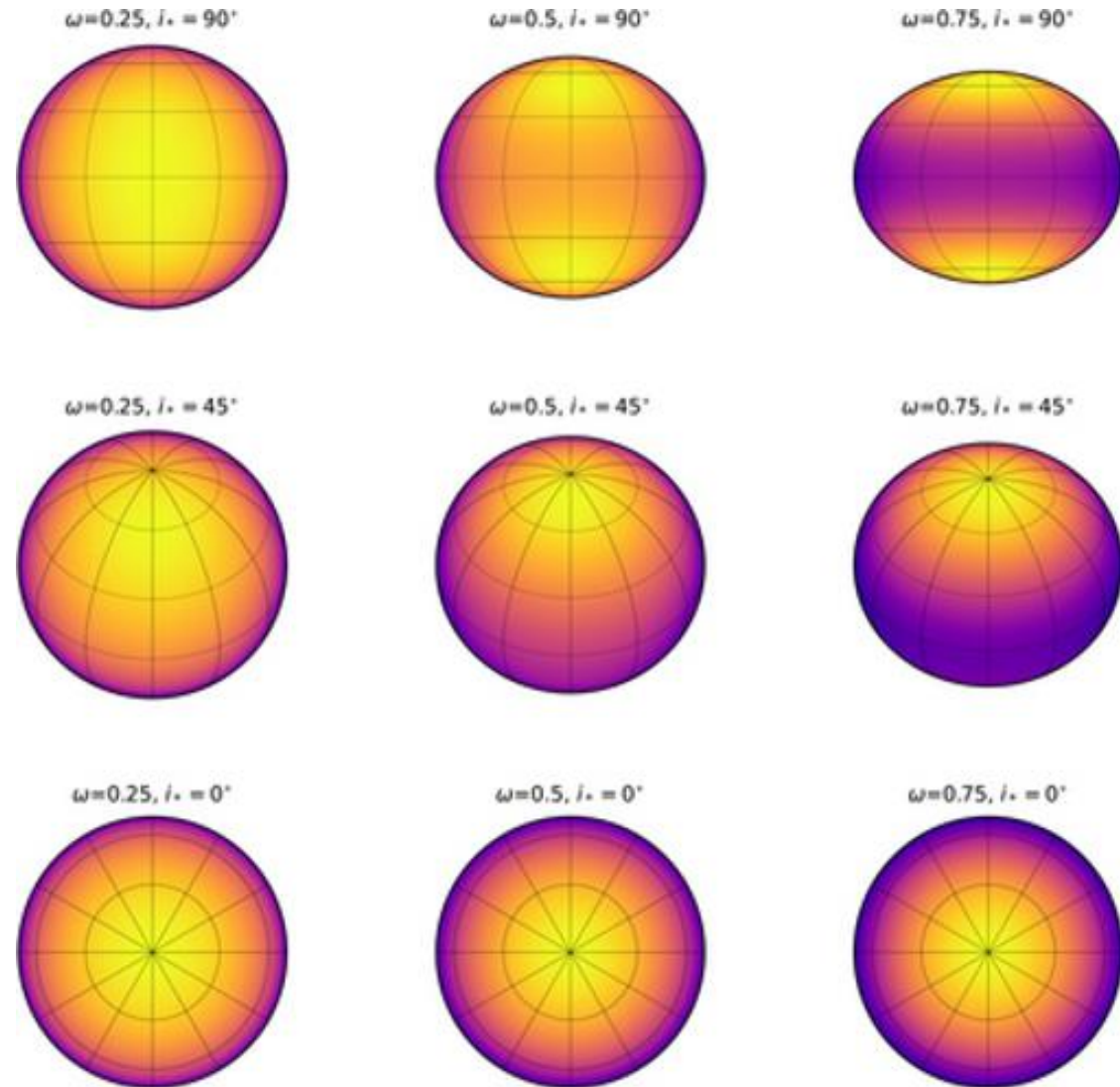
The Observational Problem

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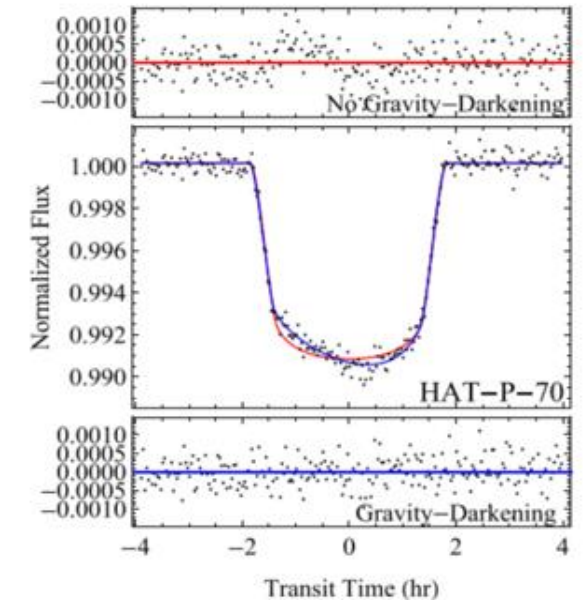
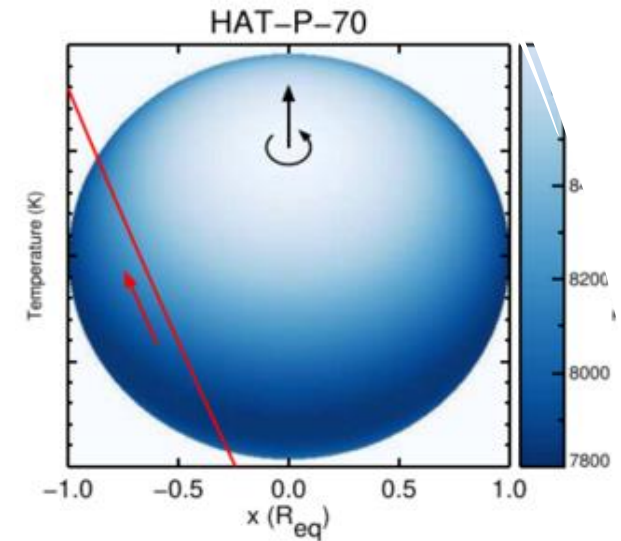
Gravity Darkening

- Rapid stellar rotation deforms the stellar surface into an oblate spheroid, producing a latitude-dependent surface gravity and temperature distribution.
- The stellar poles become hotter and brighter than the equatorial regions, generating a global brightness gradient across the stellar disk.
- When a planet transits such a star, it produces asymmetric transit light curves that encode information about the stellar inclination and orbital geometry.



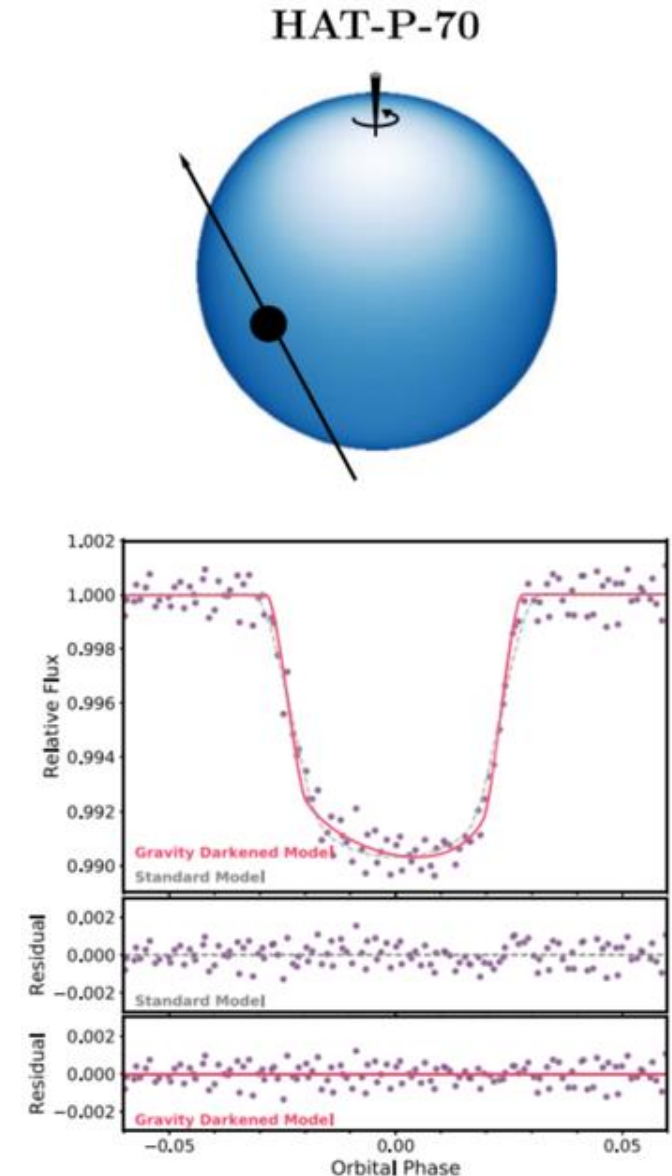
Target System: HAT-P-70 - Cauley & Ahlers, 2022

- The HAT-P-70 system was selected due to its exceptional suitability for gravity-darkening analysis.
- The host star is a rapidly rotating late A-type star with an effective temperature of approximately 8450 K, placing it well above the Kraft break, and a projected rotational velocity of $v \sin i_\star \approx 100 \text{ km s}^{-1}$.
- Previous spectroscopic studies report a strongly misaligned, retrograde orbit, with a sky-projected spin-orbit angle of $\lambda \approx 113^\circ$, an orbital inclination of $i_{\text{orb}} = 96.5^\circ$ and a stellar inclination which varies between $i_\star = 35^\circ$ and $i_\star = 58^\circ$.
- Combined with the presence of an inflated hot Jupiter on a short-period orbit, this extreme geometry maximises the observable gravity-darkening signal and makes HAT-P-70 an ideal target.

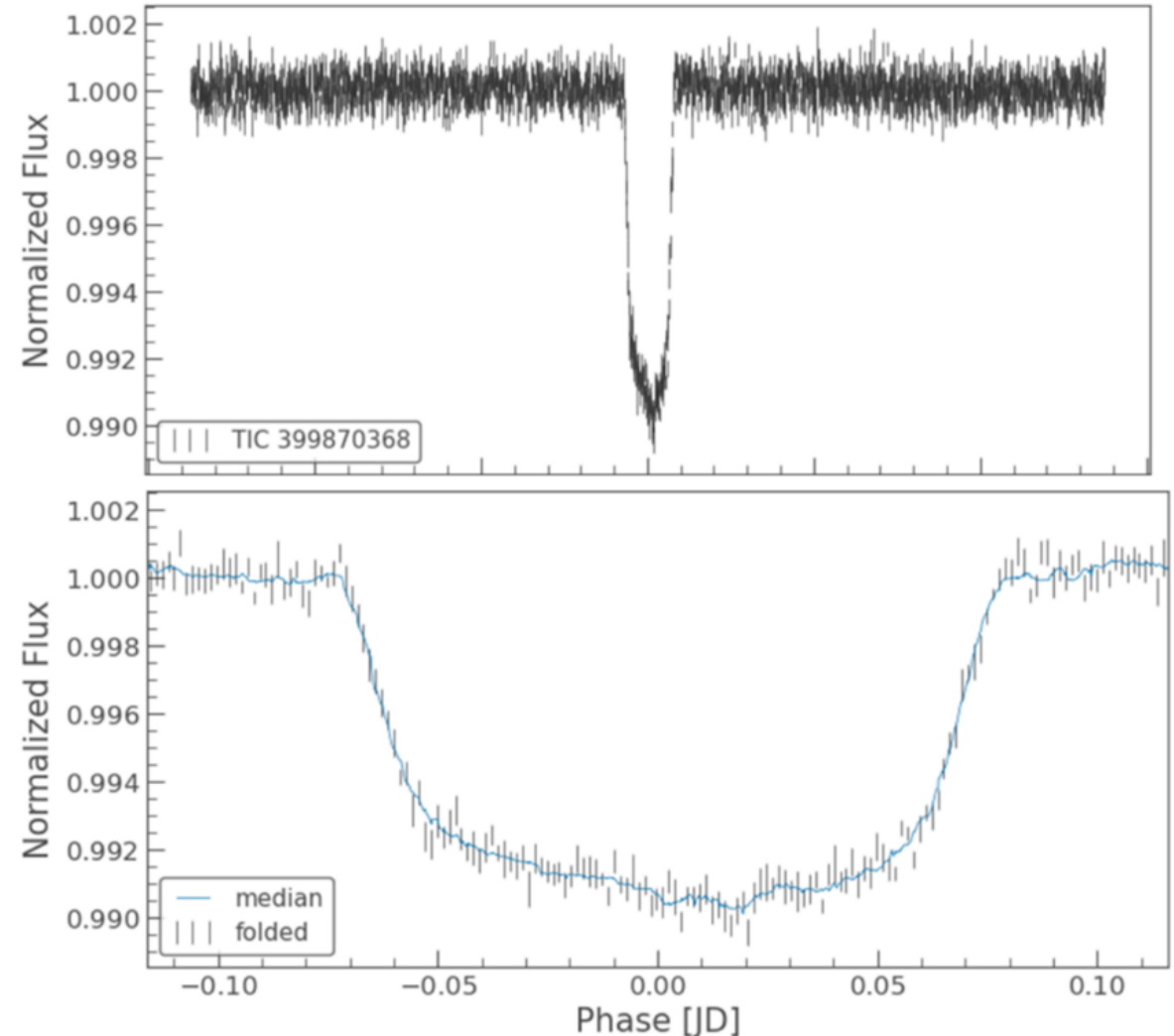


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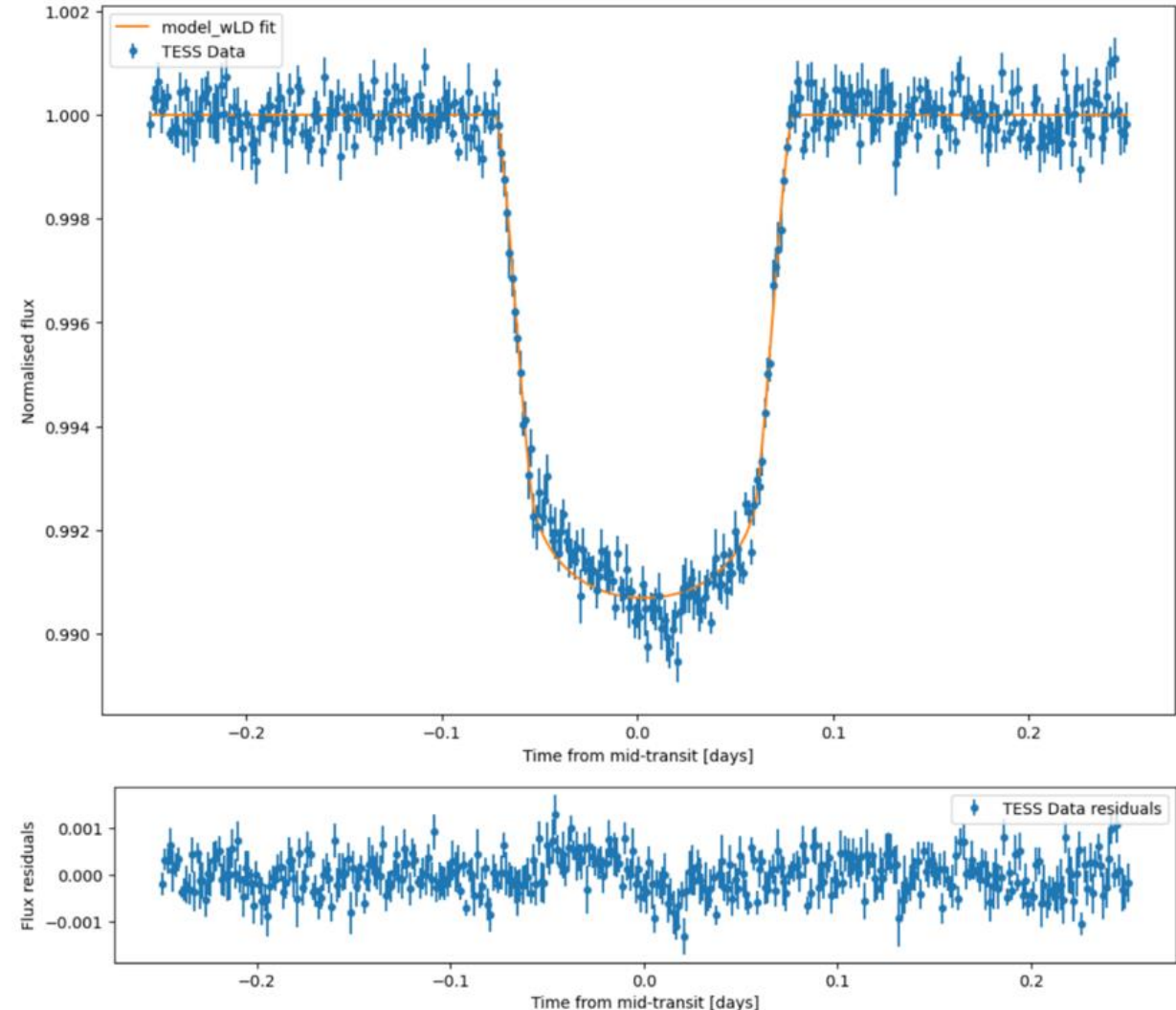


- High-precision photometry from TESS Sector 32 was used for this analysis.
- The light curves were cleaned, detrended, phase-folded, and corrected for light contamination from nearby sources.
- Processed transit revealed a statistically significant asymmetry in the light-curve profile that could not be reproduced by standard limb-darkened models alone.
- This behaviour motivated the inclusion of additional physical effects gravity darkening in the modelling framework.

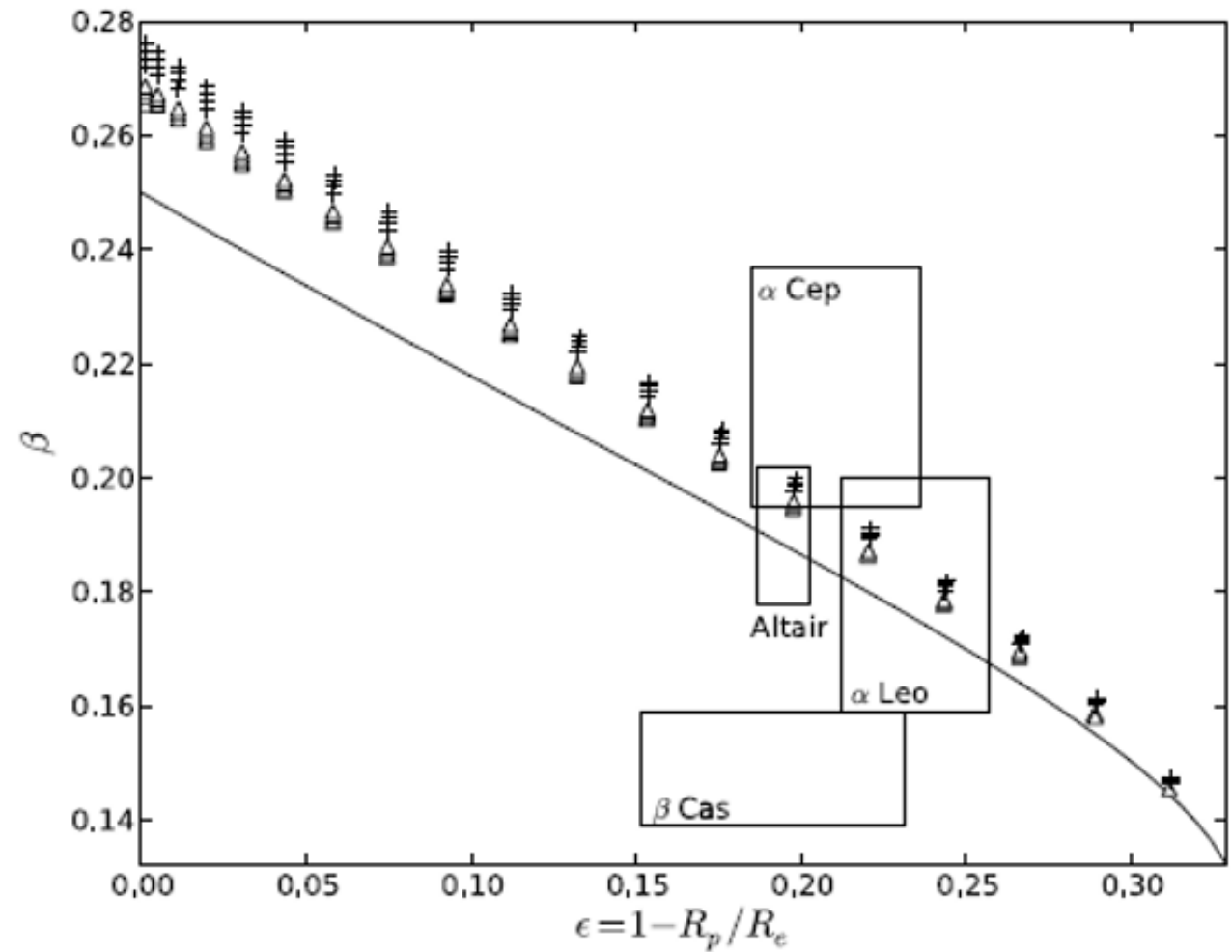


TESS Data & Initial Inspection

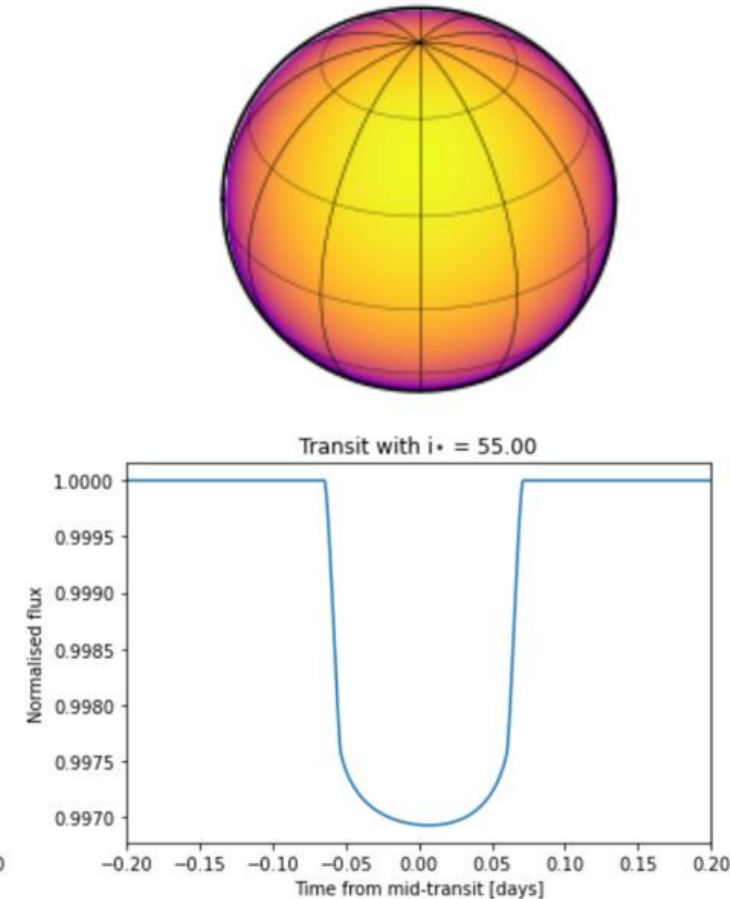
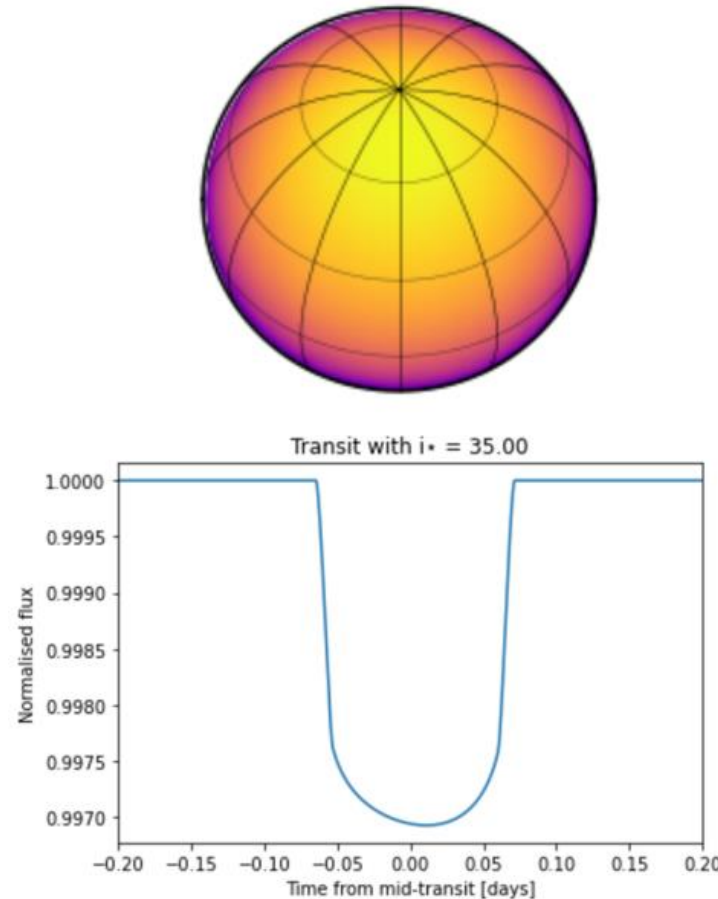
- Standard limb-darkened transit models were first applied to establish a statistical baseline.
- Fitting orbital and geometric parameters improved the overall fit systematic residuals persisted during ingress and egress.
- Statistical analysis revealed a low possibility for this to be purely by chance.
- These results indicated that a symmetric transit model was insufficient to reproduce the observed light-curve morphology.



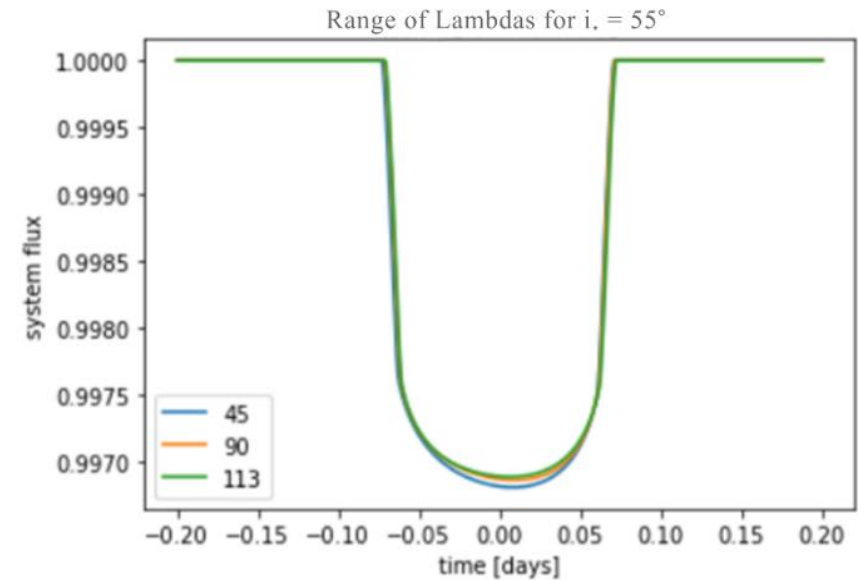
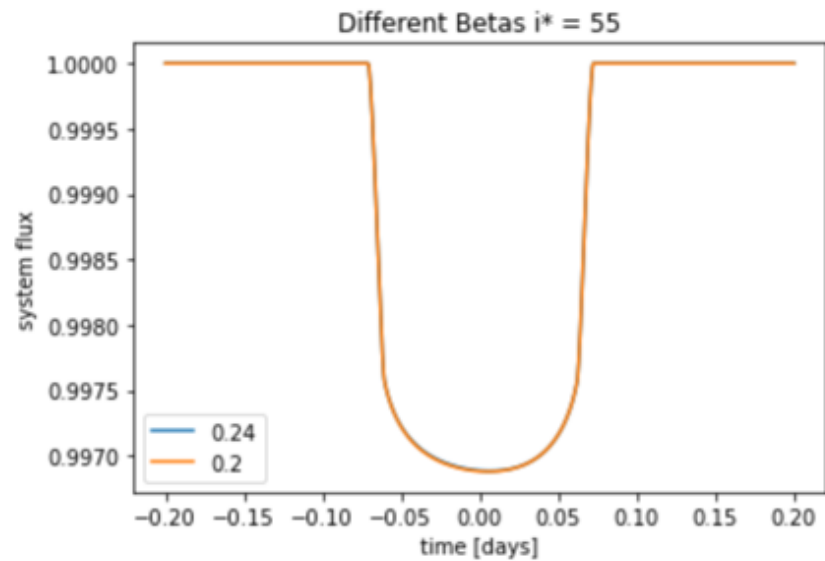
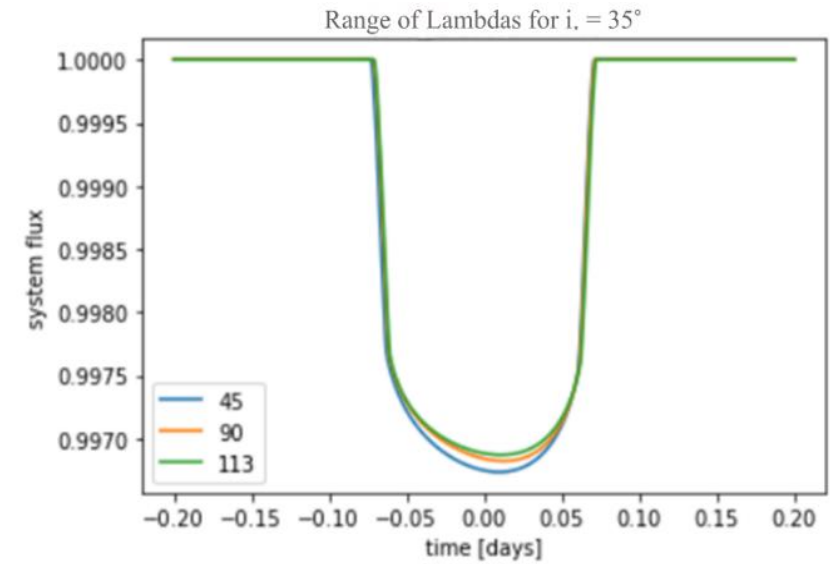
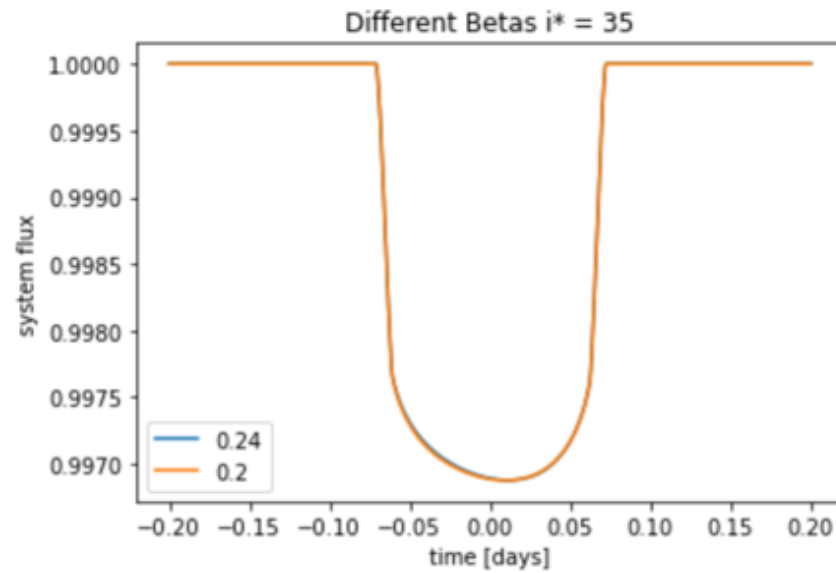
- Stellar oblateness and gravity darkening were incorporated into the transit modelling framework.
- An oblateness-dependent gravity-darkening prescription was adopted to ensure physical consistency for a rapidly rotating host star.
- Gravity-darkened transit light curves were generated using the STARRY framework, enabling the effects of stellar inclination and spin-orbit misalignment to be explored withing a self-consistent geometric model.



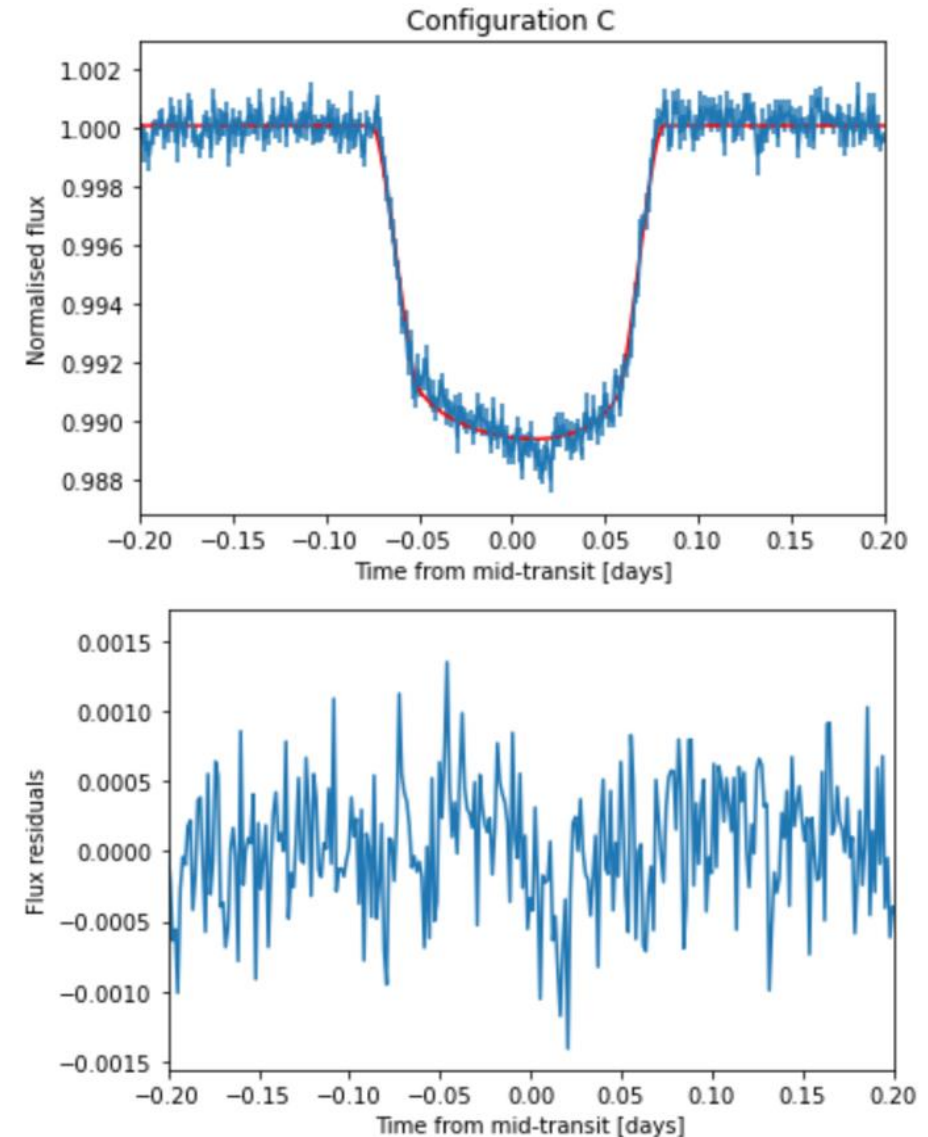
- Simulations demonstrated that multiple combinations of i_* , λ , and β parameters could reproduce similar transit morphologies.
- This degeneracy limited the ability to identify a unique best-fitting configuration from photometry alone.
- A statistical inference framework was therefore required to robustly quantify parameter uncertainties and explore correlations within the multidimensional parameter space.



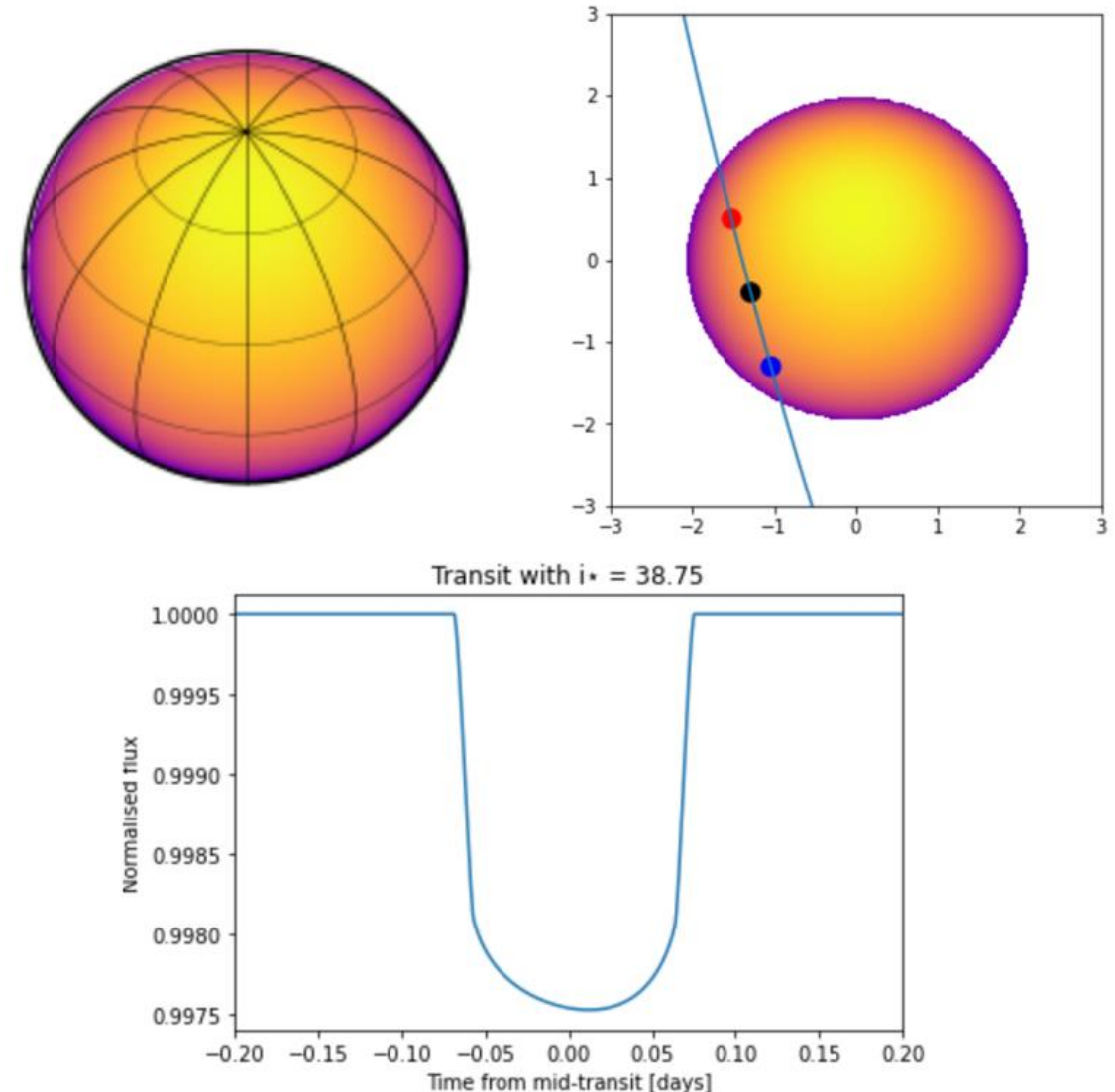
Parameters Degeneracies



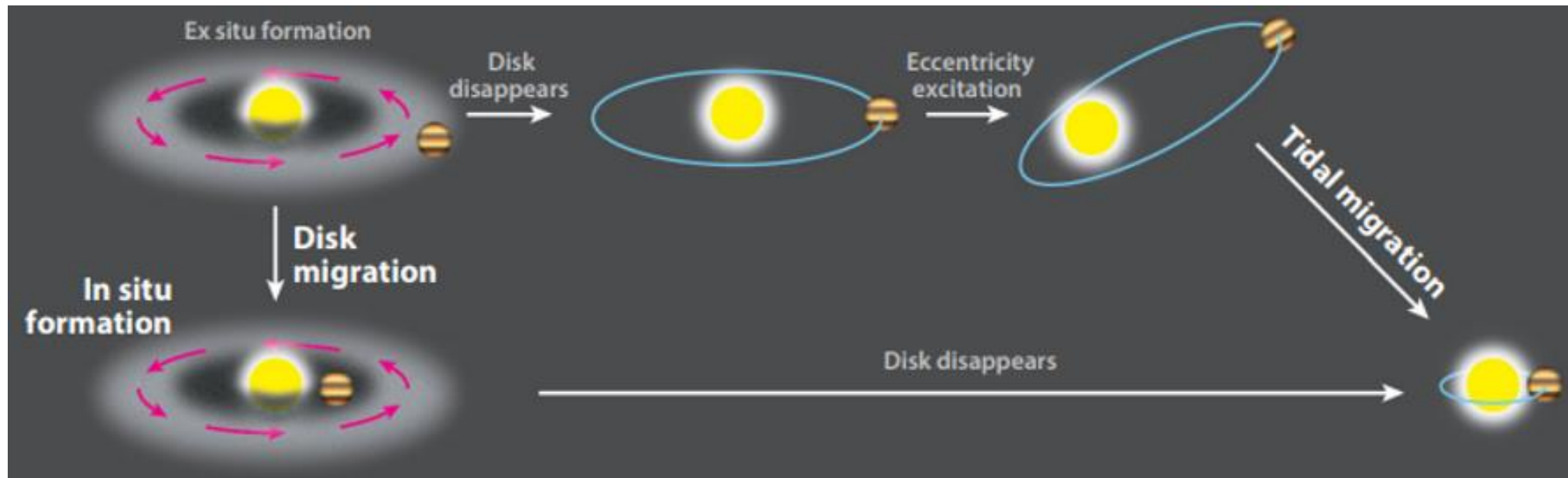
- A Markov Chain Monte Carlo (MCMC) framework was employed to explore the full parameter space of the system - $i_\star$, λ , i_{orb} , f , $R_\star$, $M_\star$
- Key Results:
 - Stellar inclination: $i_\star = 38.75 \pm 0.5^\circ$
 - Orbital inclination: $i_{\text{orb}} = 97.60 \pm 0.12^\circ$
 - Projected spin-orbit angle: $\lambda = 105.2 \pm 3.6^\circ$
 - True 3D spin-orbit angle: $\psi = 105.26^\circ$
- The inferred posteriors consistently favoured a strongly misaligned, retrograde orbital configuration, reinforcing the geometric interpretation suggested by the transit asymmetry.



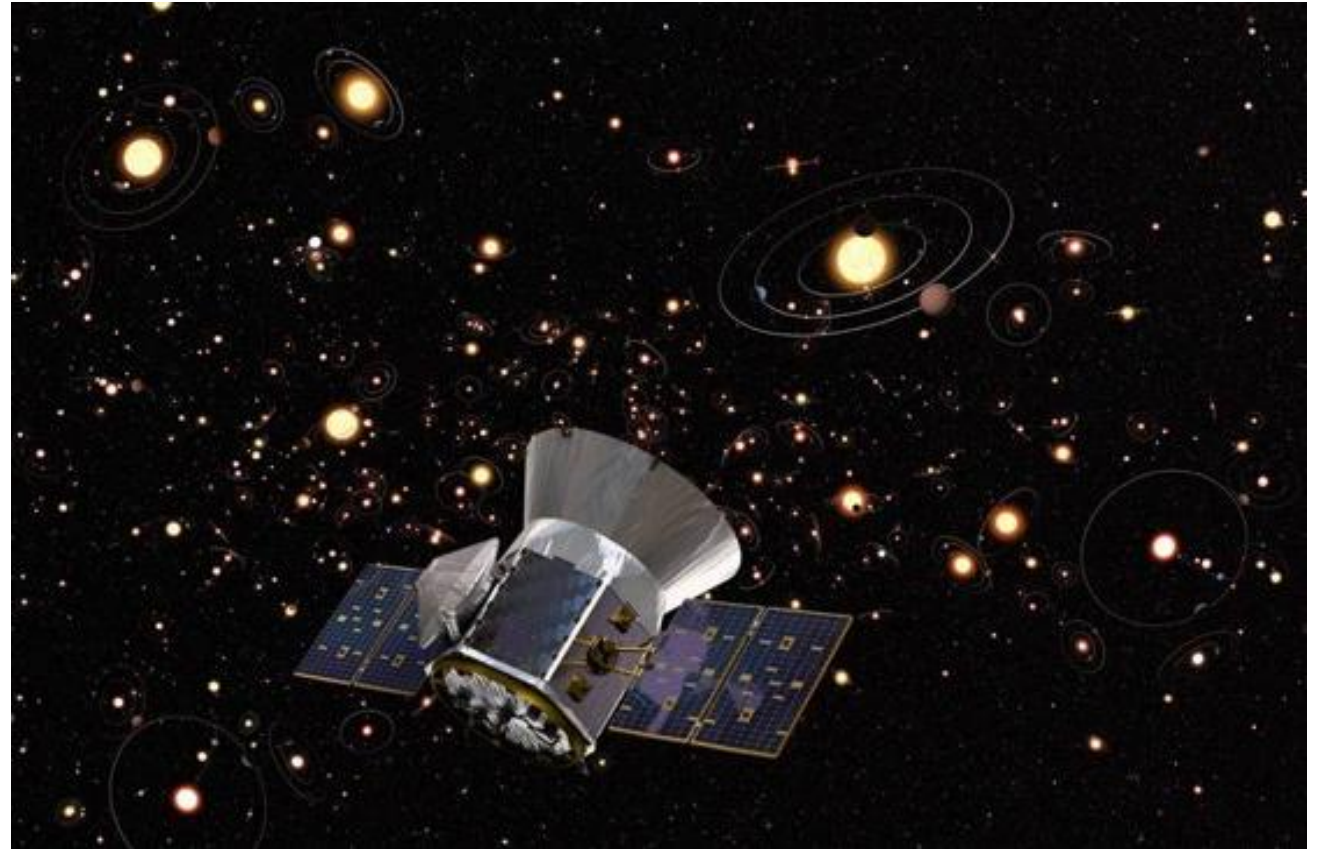
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- Spin-orbit angles provide key insight into the formation and dynamical evolution of close-in giant planets.
- In aligned systems, smooth disc-driven migration is typically favoured.
- Misaligned or retrograde configurations are often associated with high-eccentricity migration mechanisms followed by tidal circularisation.
- The misaligned, retrograde spin-orbit angle of HAT-P-70 suggested ex-situ formation followed by high-eccentricity migration.



- The precision of the inferred results remained limited by degeneracies within the gravity-darkened modelling framework and the availability of a single TESS observing sector.
- Unexplained mid-transit residual structure suggests additional physical effects (e.g., starspots, stellar pulsations) may be present but were not fully characterised.
- New TESS sectors would further refine constraints on the stellar inclination and true spin-orbit angle.



Questions?

REFERENCES



Vanderburg A. “Transit Light Curve Tutorial. <https://avanderburg.github.io/tutorial/tutorial.html> [Slide 1]

Masuda K. 2015. "Spin-Orbit Angles of Kepler-13Ab and HAT-P-7b from Gravity-Darkened Transit Light Curves." The Astrophysical Journal, volume 805, number 1, article 28, <https://doi.org/10.1088/0004-637X/805/1/28> [Slide 3]

Gaudi B. S. & Winn J. N. 2007. "Prospects for the Characterization and Confirmation of Transiting Exoplanets via the Rossiter-McLaughlin Effect." The Astrophysical Journal, volume 655, number 1, article 550, <https://doi.org/10.1086/509910> [Slide 4]

Johnson M. C. et al. 2017. "Spin-Orbit Misalignments of Three Jovian Planets via Doppler Tomography." The Astronomical Journal, volume 154, number 4, article 137, <https://doi.org/10.3847/1538-3881/aa8462> [Slide 5]

Dholakia S. et al. 2022. “Efficient and Precise Transit Light Curves for Rapidly Rotating, Oblate Stars.” The Astrophysical Journal, volume 925, number 2, article 185, <https://doi.org/10.3847/1538-4357/ac33aa> [Slide 6]

Cauley P. W. & Ahlers J. P. 2022. “The Effects of Stellar Gravity Darkening on High-resolution Transmission Spectra.” The Astronomical Journal, volume 163, number 3, article 122, <https://doi.org/10.3847/1538-3881/ac478f> [Slide 7]

REFERENCES



Zhou G. et al. 2019. “Two New HATNet Hot Jupiters around A Stars and the First Glimpse at the Occurrence Rate of Hot Jupiters from TESS.” The Astronomical Journal, volume 158, number 4, article 141, <https://doi.org/10.3847/1538-3881/ab36b5> [Slide 8]

Espinosa F. L. & Rieutord M. 2011. “Gravity darkening in rotating stars.” Astronomy & Astrophysics, volume 533, article 43, <https://doi.org/10.1051/0004-6361/201117252> [Slide 11]

Dawson R. I. & Johnson J. A. 2018. “Origins of hot Jupiters.” Annual Review of Astronomy and Astrophysics, volume 56, number 1, pages 175-221, <https://doi.org/10.1146/annurev-astro-081817-051853> [Slide 16]

News Staff 2019. “TESS Astronomers Create Catalog of Habitable Zone Stars.” Sci News, <https://www.sci.news/astronomy/tess-catalog-habitable-zone-stars-07034.html> [Slide 17]